

## TARGOVISTE, Romania

WMO: 153750

Lat: 44.9295N

Lon: 25.4263E

Elev: 977

StdP: 14.18

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 10.0       | 15.0 | 5.3                           | 7.5 | 11.8 | 10.0 | 9.5 | 17.3 | 26.1                  | 36.4 | 17.8 | 38.0 | 4.3                   | 320  | 0.468 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 22.0                   | 89.6            | 69.2 | 87.3 | 68.4 | 84.8 | 67.6 | 72.6                | 84.2 | 71.0 | 82.1 | 69.7 | 80.5 | 4.9                  | 70   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 69.2                            | 111.2 | 76.8 | 67.4 | 104.6 | 75.3 | 66.1 | 99.6 | 74.3 | 36.8          | 84.2 | 35.4 | 82.2 | 34.3 | 80.6 | 79.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       |
| Min               | Max  |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |
| 15.9              | 12.8 | 10.6 | 1.5                        | 94.3 | 5.1                | 4.0 | -2.1                                               | 97.2 | -5.1       | 99.5 | -8.0       | 101.8 | -11.7      | 104.7 |
|                   |      |      | 1.2                        | 77.1 | 5.1                | 2.8 | -2.5                                               | 79.1 | -5.5       | 80.7 | -8.3       | 82.2  | -12.0      | 84.2  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |       |            |       |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |       |            |       |

## Monthly Climatic Design Conditions

|                                                                                  |                                                                                  | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                                  |                                                                                  | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg                                                                            | 52.0   | 31.0  | 35.8  | 42.1  | 51.6  | 60.9  | 68.7  | 70.8  | 71.3  | 62.9  | 52.3  | 41.6  | 33.8 |      |
|                                                                                  | DBStd                                                                            | 15.89  | 8.23  | 8.55  | 7.75  | 7.28  | 5.87  | 4.80  | 4.69  | 4.90  | 7.11  | 7.07  | 8.02  | 8.23 |      |
|                                                                                  | HDD50                                                                            | 2154   | 589   | 399   | 259   | 68    | 4     | 0     | 0     | 0     | 1     | 58    | 273   | 503  |      |
|                                                                                  | HDD65                                                                            | 5362   | 1054  | 816   | 710   | 403   | 151   | 20    | 10    | 9     | 123   | 396   | 703   | 967  |      |
|                                                                                  | CDD50                                                                            | 2874   | 0     | 3     | 14    | 116   | 341   | 560   | 644   | 659   | 388   | 129   | 20    | 0    |      |
|                                                                                  | CDD65                                                                            | 608    | 0     | 0     | 0     | 1     | 23    | 130   | 189   | 203   | 59    | 3     | 0     | 0    |      |
|                                                                                  | CDH74                                                                            | 6203   | 0     | 0     | 0     | 53    | 312   | 1234  | 1809  | 2061  | 684   | 50    | 0     | 0    |      |
|                                                                                  | CDH80                                                                            | 2008   | 0     | 0     | 0     | 4     | 52    | 340   | 603   | 814   | 192   | 3     | 0     | 0    |      |
| Wind                                                                             | WSAvg                                                                            | 4.7    | 3.9   | 4.5   | 5.1   | 5.6   | 5.4   | 4.9   | 4.8   | 4.9   | 5.1   | 4.5   | 4.0   | 4.0  |      |
| Precipitation                                                                    | PrecAvg                                                                          | 26.9   | 1.5   | 1.2   | 1.5   | 2.3   | 3.1   | 3.6   | 3.3   | 2.3   | 2.4   | 2.1   | 1.7   | 1.8  |      |
|                                                                                  | PrecMax                                                                          | 41.2   | 2.8   | 3.1   | 3.7   | 6.3   | 6.5   | 7.8   | 7.9   | 9.9   | 5.8   | 4.1   | 3.9   | 5.7  |      |
|                                                                                  | PrecMin                                                                          | 14.8   | 0.1   | 0.1   | 0.0   | 0.8   | 0.6   | 1.2   | 1.1   | 0.2   | 0.3   | 0.0   | 0.1   | 0.0  |      |
|                                                                                  | PrecStd                                                                          | 6.2    | 0.8   | 0.8   | 1.0   | 1.4   | 1.5   | 1.5   | 1.9   | 1.9   | 1.5   | 1.2   | 1.1   | 1.3  |      |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                             | DB     | 56.5  | 66.0  | 70.5  | 79.6  | 85.2  | 90.1  | 92.3  | 94.2  | 89.8  | 79.0  | 68.3  | 60.1 |      |
|                                                                                  |                                                                                  | MCWB   | 46.9  | 52.0  | 53.0  | 60.6  | 65.8  | 71.1  | 71.5  | 70.4  | 66.6  | 62.6  | 55.7  | 48.9 |      |
|                                                                                  | 2%                                                                               | DB     | 49.5  | 58.9  | 64.9  | 74.4  | 80.6  | 86.9  | 88.7  | 90.0  | 85.0  | 74.1  | 63.1  | 53.5 |      |
|                                                                                  |                                                                                  | MCWB   | 42.7  | 48.3  | 50.9  | 58.5  | 64.3  | 69.6  | 70.4  | 68.3  | 64.8  | 60.8  | 53.4  | 46.2 |      |
|                                                                                  | 5%                                                                               | DB     | 45.7  | 53.5  | 60.5  | 69.6  | 77.3  | 83.9  | 86.0  | 87.7  | 81.5  | 69.7  | 57.7  | 49.1 |      |
|                                                                                  |                                                                                  | MCWB   | 40.9  | 45.5  | 48.7  | 55.7  | 62.6  | 68.0  | 69.2  | 67.6  | 64.2  | 58.0  | 51.0  | 44.3 |      |
|                                                                                  | 10%                                                                              | DB     | 41.4  | 48.5  | 56.3  | 65.7  | 74.0  | 81.2  | 83.4  | 84.9  | 77.7  | 64.9  | 53.1  | 44.5 |      |
|                                                                                  |                                                                                  | MCWB   | 38.4  | 42.6  | 46.5  | 54.2  | 61.0  | 66.6  | 68.2  | 67.3  | 63.0  | 55.6  | 48.3  | 41.0 |      |
|                                                                                  | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 47.6  | 53.2  | 54.8  | 62.1  | 68.6  | 74.2  | 75.4  | 74.5  | 70.2  | 66.5  | 57.0 | 50.1 |
|                                                                                  |                                                                                  |        | MCDB  | 55.7  | 64.4  | 68.4  | 76.4  | 79.9  | 85.4  | 87.2  | 86.7  | 82.0  | 75.1  | 65.0 | 55.1 |
| 2%                                                                               |                                                                                  | WB     | 44.1  | 49.2  | 51.7  | 59.5  | 66.0  | 71.8  | 72.7  | 71.8  | 68.1  | 61.9  | 54.6  | 47.5 |      |
|                                                                                  |                                                                                  | MCDB   | 47.8  | 57.6  | 62.9  | 72.2  | 77.7  | 82.9  | 84.3  | 83.7  | 79.3  | 69.9  | 62.3  | 52.8 |      |
| 5%                                                                               |                                                                                  | WB     | 41.3  | 46.1  | 49.4  | 57.2  | 63.9  | 70.0  | 71.1  | 70.2  | 65.8  | 59.1  | 51.8  | 44.3 |      |
|                                                                                  |                                                                                  | MCDB   | 45.0  | 53.1  | 59.3  | 67.5  | 74.6  | 81.1  | 81.9  | 81.6  | 76.2  | 68.5  | 56.4  | 48.9 |      |
| 10%                                                                              |                                                                                  | WB     | 38.7  | 43.3  | 47.3  | 55.1  | 62.1  | 68.3  | 69.8  | 68.8  | 63.9  | 56.8  | 49.2  | 41.3 |      |
|                                                                                  |                                                                                  | MCDB   | 41.1  | 47.4  | 55.0  | 63.5  | 71.4  | 78.6  | 79.9  | 80.0  | 74.0  | 63.5  | 52.6  | 44.0 |      |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB                                                                            | MDBR   | 15.2  | 17.0  | 18.7  | 20.2  | 20.5  | 21.1  | 22.0  | 22.9  | 21.5  | 18.9  | 14.9  | 15.5 |      |
|                                                                                  |                                                                                  | MCDBR  | 20.7  | 26.2  | 27.6  | 28.6  | 26.4  | 25.8  | 26.3  | 27.5  | 27.9  | 26.9  | 23.9  | 22.3 |      |
|                                                                                  |                                                                                  | MCWBR  | 13.9  | 16.4  | 15.6  | 14.8  | 12.5  | 11.5  | 10.5  | 9.8   | 10.4  | 13.8  | 14.0  | 14.4 |      |
|                                                                                  | 5% WB                                                                            | MCDBR  | 18.9  | 25.1  | 25.8  | 25.6  | 24.7  | 23.9  | 23.6  | 23.7  | 24.2  | 24.9  | 20.8  | 21.0 |      |
|                                                                                  |                                                                                  | MCWBR  | 13.2  | 16.0  | 15.1  | 13.8  | 12.7  | 11.6  | 11.1  | 10.3  | 10.1  | 13.4  | 12.5  | 13.9 |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub                                                                             | 0.327  | 0.359 | 0.399 | 0.457 | 0.433 | 0.438 | 0.450 | 0.468 | 0.423 | 0.386 | 0.359 | 0.330 |      |      |
|                                                                                  | taud                                                                             | 2.315  | 2.229 | 2.181 | 2.042 | 2.182 | 2.207 | 2.184 | 2.124 | 2.216 | 2.302 | 2.326 | 2.336 |      |      |
|                                                                                  | Ebn at Noon                                                                      | 244    | 256   | 261   | 255   | 266   | 265   | 260   | 249   | 251   | 244   | 229   | 230   |      |      |
|                                                                                  | Edh at Noon                                                                      | 26     | 34    | 40    | 50    | 45    | 44    | 45    | 46    | 39    | 31    | 26    | 23    |      |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg                                                                           | 444    | 705   | 1075  | 1437  | 1765  | 1918  | 1944  | 1738  | 1267  | 846   | 479   | 363   |      |      |
|                                                                                  | RadStd                                                                           | 57     | 100   | 115   | 150   | 151   | 151   | 143   | 121   | 108   | 106   | 56    | 51    |      |      |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) Regional (72 neighbors) |  | +0.92 | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | -265        | +126  | +54   |       |

Nomenclature: See separate page